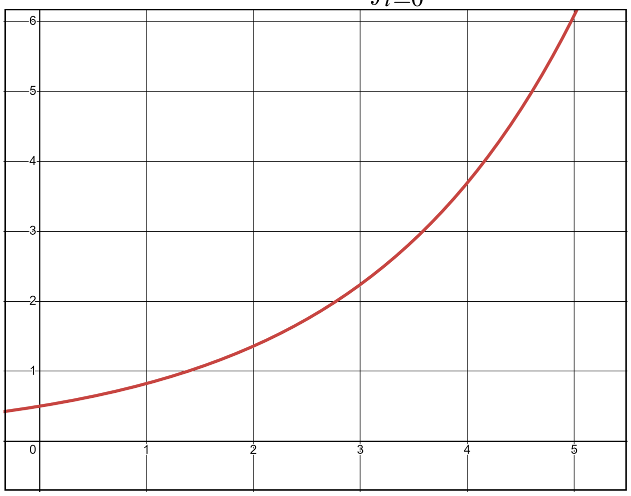
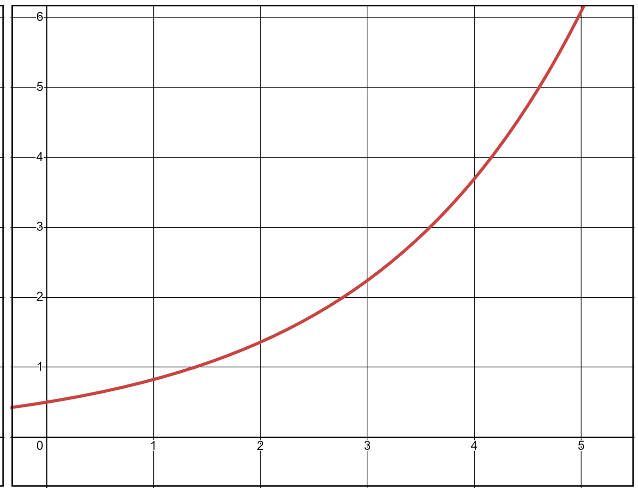
Learning target IN2 and INx, version 1

The rate at which pollution escapes a scrubbing process at a manufacturing plant increases over time as filters and other technologies become less effective. Suppose that the rate of pollution (measured in tons per week) is given by the function r that is pictured below. Throughout this problem, we're interested in $\int_{t=0}^{5} r(t)\,dt$.





(a) On the first graph above, carefully draw the *left* Riemann sum with four rectagles of uniform width. (How wide should each rectangle be?)

(b) On the second graph above, carefully draw the *right* Riemann sum with four rectangles.

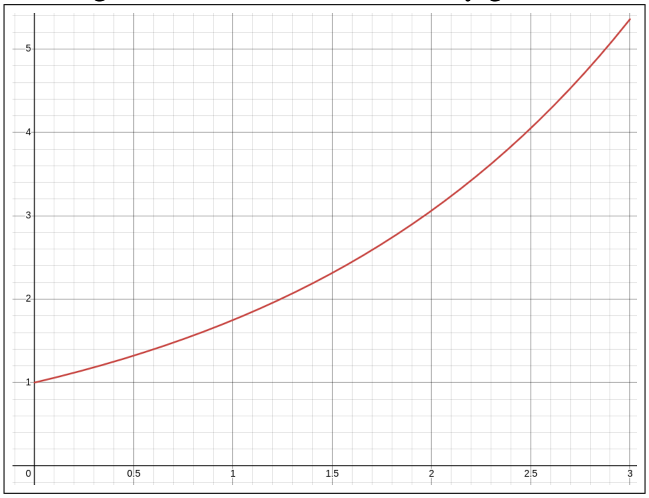
(c) If $r(t) = 0.5e^{0.5t}$, compute both of these Riemann sums. Show your work for computing the heights of the rectangles. (Hint: lots of your work will overlap!)

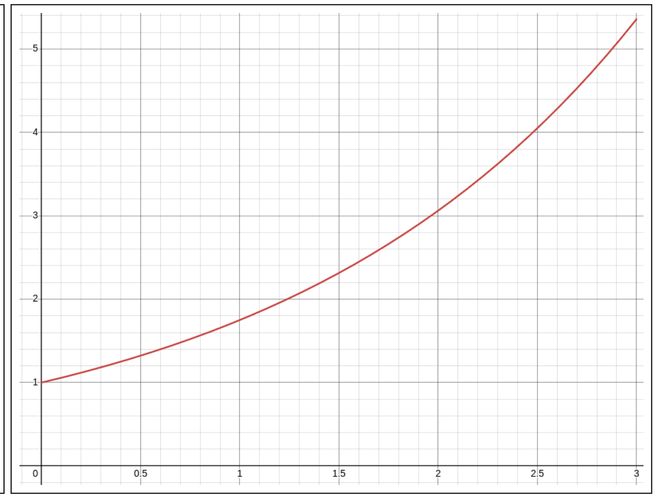
Include units on every number you write.

(d) Give your best guess for $\int_{t=0}^{5} r(t)\,dt$, **include units**, and explain what this number means about pollution.

Learning target IN2 and INx, version 2

A colony of bacteria in a petri dish is growing at a rate of $f(t) = 1.75^t$ million bacteria per hour. Let's figure out how much the colony grows over the course of three hours.





(a) On the first graph above, carefully draw the *left* Riemann sum with four rectagles of uniform width. (How wide should each rectangle be?)

(b) On the second graph above, carefully draw the *right* Riemann sum with four rectangles.

(c) Compute both of these Riemann sums. Show your work for computing the heights of the rectangles. (Hint: lots of your work will overlap!)

Include units on every number you write.

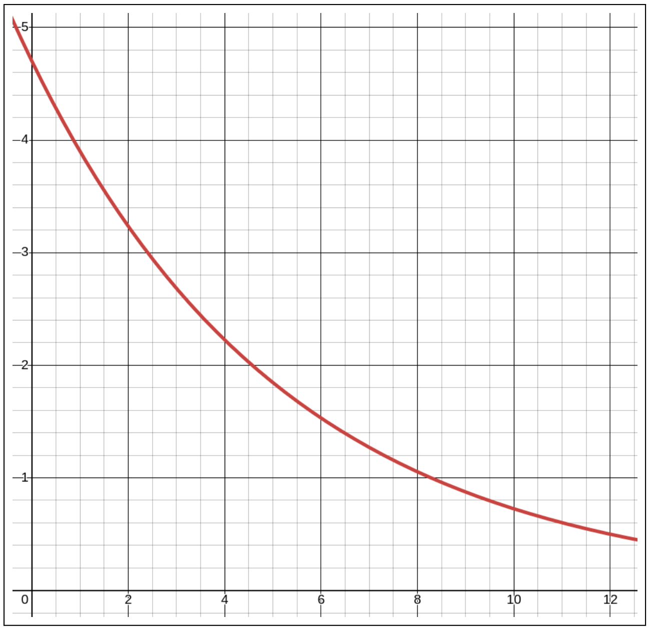
(d) Give your best guess for $\int_{t=0}^{3} f(t)\,dt$, **include units**, and explain what this number means about the bacteria colony.

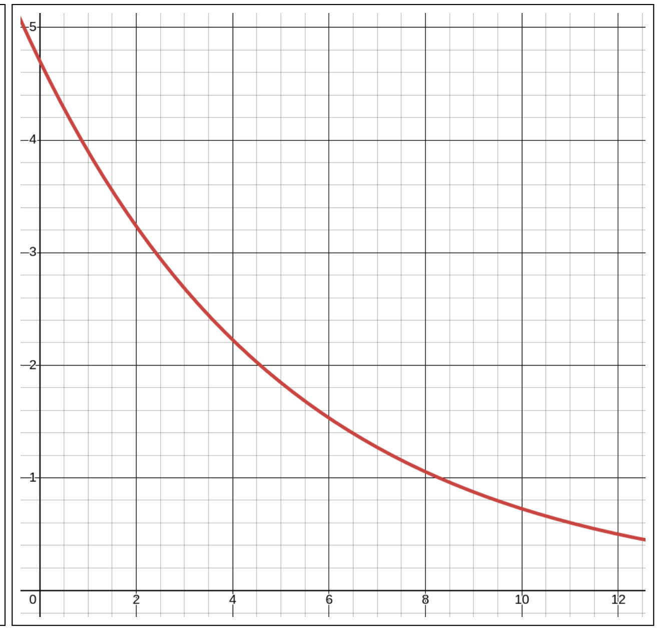
MATH 201 Your name: ______________________________

Learning target IN2 and INx, version 3

Over the first year of its life, a golden retriever puppy gains weight at a rate of

$$p(t) = 4.7e^{-0.187t} \text{ kilograms per month.}$$





(a) On the first graph above, carefully draw the *left* Riemann sum with four rectagles of uniform width. (How wide should each rectangle be?)

(b) On the second graph above, carefully draw the *right* Riemann sum with four rectangles.

(c) Compute both of these Riemann sums. Show your work for computing the heights of the rectangles. (Hint: lots of your work will overlap!)

Include units on every number you write.

(d) Give your best guess for $\displaystyle\int_{t=0}^{12} p(t)\,dt$, **include units**, and explain what this number means about the puppy.

Learning target INx, version 3

Over the first year of its life, a golden retriever puppy gains weight at a rate of

$$p(t) = 4.7e^{-0.187t} \text{ kilograms per month.}$$

(a) On the graph above, carefully draw a Riemann sum with four rectangles of uniform width. (How wide should each rectangle be?)

(b) Choose one of the Riemann rectangles you drew and shade it in for clarity. For this rectangle, compute and label:

the height: $\underbrace{\qquad\qquad}_{\text{number}}\ \underbrace{\qquad}_{\text{units}}$ the width: $\underbrace{\qquad\qquad}_{\text{number}}\ \underbrace{\qquad}_{\text{units}}$ the area: $\underbrace{\qquad\qquad}_{\text{number}}\ \underbrace{\qquad}_{\text{units}}$

(c) What would the overall area $\int_{t=0}^{12} p(t)\,dt$ mean about the puppy? **Give units in your answer.**

MATH 201 Your name: ____________________

Learning target INx, version 4

Suppose that an underground storage tank leaks pollutants at a rate of $r(t)$ gallons per day, where t is measured in days. A researcher studying watershed quality develops the model graphed below:

$$r(t) = 0.0069t^3 - 0.14t^2 + 11.485.$$

The researcher is interested in $\int_{t=2}^{10} r(t)\, dt$.

(a) On the graph above, carefully draw a Riemann sum between $t = 2$ and $t = 10$ with four rectangles of uniform width.
(How wide should each rectangle be?)

(b) Choose one of the Riemann rectangles you drew and shade it in for clarity. For this rectangle, compute and label:

the height: $\underbrace{\qquad\quad}_{\text{number}}\ \underbrace{\qquad}_{\text{units}}$ the width: $\underbrace{\qquad\quad}_{\text{number}}\ \underbrace{\qquad}_{\text{units}}$ the area: $\underbrace{\qquad\quad}_{\text{number}}\ \underbrace{\qquad}_{\text{units}}$

(c) What would the overall area $\int_{t=2}^{10} r(t)\, dt$ mean about the watershed? **Give units!**